

Real-Time Privacy Guardianship in XR Metaverses via AI-Driven Behavioral Analytics

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Abstract—The metaverse introduces embodied presence through avatars, creating novel privacy vulnerabilities in which behavioral patterns become primary attack surfaces for tracking, impersonation, and interaction profiling. This paper presents BehaviorShield, a real-time privacy intrusion detection system grounded in behavioral analytics applied to genuine XR motion data. The system is trained on the WhoIsAlyx VR motion dataset (9 players, 596,182 benign frames), augmented with 119,236 frames from the Project Aria Everyday Activities (AEA) dataset. Three simulated threat types, avatar tracking, behavioral impersonation, and spatial profiling, yield 993,636 annotated frames. A Random Forest classifier trained on 19 features with SMOTE resampling achieves a mean ROC-AUC of 0.9673 ± 0.0089 under player-level Leave-One-Subject-Out (LOSO) cross-validation and 0.9759 AUC on the best held-out fold. SHAP-based explainability identifies rolling motion variability as the dominant discriminator across all threat scenarios.

Index Terms—metaverse security, behavioral analytics, random forest, VR motion data, privacy protection, avatar tracking, LOSO cross-validation, SHAP explainability.

I. INTRODUCTION

The metaverse [1]–[3] is emerging as the next evolutionary stage of the internet, a persistent, immersive virtual world where users interact through digital avatars in real time. Enabled by VR, AR, 5G/6G [4], and blockchain [5]–[7], it introduces embodied presence, world persistence, and native digital economies. These properties, however, give rise to severe privacy threats. In the metaverse, avatar movement is continuous, high-fidelity, and semantically rich, making behavioral patterns primary attack surfaces [8], [9]. Three dominant threat classes have been identified:

- **Avatar Tracking:** Adversaries monitor avatar movements to infer real-world locations and routines.
- **Behavioral Impersonation:** Attackers mimic the motion signatures of trusted users to deceive social and security systems.
- **Spatial Profiling:** Systematic traversal of virtual spaces to construct behavioral dossiers mapping avatar habits.

Conventional security mechanisms [10], [11] are ill-suited to these threats. Encryption cannot prevent behavioral inference from observable avatar kinematics; access control cannot anticipate dynamic avatar interactions; signature-based detection fails against novel behavioral attack patterns [12].

This paper presents BehaviorShield, a behavioral analytics pipeline for real-time privacy threat detection. Built on genuine XR motion capture data from real VR players, it contributes: (1) a 9-player WhoIsAlyx dataset with 3 simulated threat types (993,636 annotated samples); (2) AEA cross-domain augmentation (119,236 real-world frames per fold); (3) 19-feature engineering, including 12 rolling 3-second statistics; (4) rigorous LOSO evaluation at both player and date levels; (5) SHAP-based per-scenario explainability; and (6) adaptive adversary evasion robustness analysis.

The remainder of this paper is organized as follows. Section II reviews related work on metaverse security and behavioral anomaly detection. Section III describes the BehaviorShield system architecture. Section IV details the dataset construction, threat simulation, and evaluation protocol. Section V presents experimental results. Section VI discusses key insights, limitations, and future directions, and Section VII concludes the paper.

II. RELATED WORK

The metaverse merges physical and digital realities through avatar embodiment and persistent virtual presence, introducing behavioral privacy vulnerabilities absent in traditional web systems [13], [14]. Prior work identifies avatar-layer attacks as an emerging threat class [15], with risks arising from continuous motion data and identity-linked interactions [8]. In particular, movement tracking can reveal real-world identity through gait-like patterns, while behavioral mimicry exploits trust embedded in familiar motion signatures [16].

Machine learning approaches have been proposed for metaverse [17] intrusion detection, often relying on syn-

thetic behavioral data and limited cross-user evaluation [18]. Other works explore blockchain-assisted detection [19] and deepfake-driven impersonation attacks [20], highlighting the growing need for robust behavioral analysis.

Recent avatar-centric anomaly detection methods include gaze tracking [21], movement clustering [22], and interaction graph modeling [23]. However, these approaches share key limitations: (i) reliance on synthetic data lacking biomechanical realism, (ii) evaluation on same-session samples that inflate performance, and (iii) lack of explainability. Additionally, real-time metaverse systems require sub-10ms inference at 90fps VR rates. BehaviorShield addresses these challenges through real XR motion data, rigorous LOSO evaluation, and interpretable SHAP-based analysis [24].

III. SYSTEM ARCHITECTURE

BehaviorShield employs a three-tier architecture for real-time behavioral privacy threat detection.

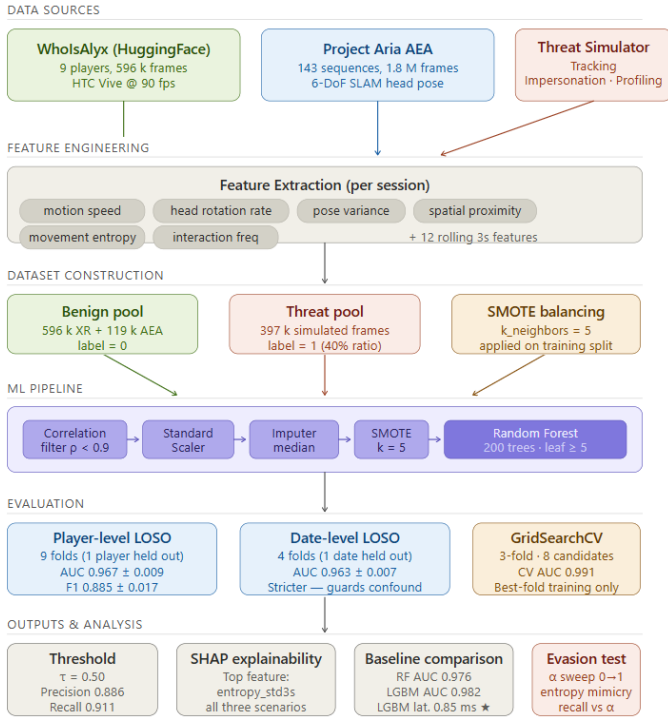


Fig. 1. BehaviorShield three-tier architecture. Raw VR head-pose frames enter the Data Ingestion Layer; 19 behavioral features are extracted in the Feature Engineering Layer; the RF classifier with SMOTE, SHAP explainer, and threshold module operate in the Detection Layer.

A. Data Ingestion Layer

The system ingests head-pose trajectories from VIVE VR motion capture at 90fps. Each frame delivers seven-dimensional state vectors: head position (x, y, z), head rotation quaternion (q_x, q_y, q_z, q_w), and a millisecond-precision timestamp. The WhoIsAlyx dataset provides 9-player data across 4 recording sessions (596,182 benign frames, ≈ 97 minutes), spanning 2022-01-07 to 2022-01-17. Project Aria AEA sequences [25] provide real-world egocentric head motion at

90fps (143 sequences) as out-of-distribution benign augmentation, confirmed as domain-distinct by KS-test ($p < 0.001$ across all 6 features, mean JS divergence = 0.2399) [26].

B. Feature Extraction and Engineering Layer

Six biomechanical XR features are computed per frame via kinematic differencing, quaternion geometry, and rolling statistics (Table I). A temporal feature *time_since_last* captures inter-frame timing. Rolling 3-second mean and standard deviation statistics are computed for each of the six base features, yielding 19 total features.

TABLE I
EXTRACTED XR BEHAVIORAL FEATURES

Feature	Description	Window
motion_speed	Normalized Euclidean displacement per frame	Frame
head_rotation_rate	Angular velocity from quaternion change	Frame
pose_variance	Std. dev. of position magnitude	30 frames
spatial_proximity	Inverse distance from session centroid	Frame
interaction_freq	Zero-crossing rate of motion speed	90 frames
movement_entropy	Spatial entropy over discretized grid	270 frames

C. Detection and Analytics Layer

Correlation Filtering: Features with pairwise Pearson correlation exceeding 0.9 are flagged for removal. The maximum observed correlation was $\rho = 0.9138$ (*motion_speed_mean3s* vs. *head_rotation_rate_mean3s*); one feature is dropped, yielding 18 classifier inputs.

Class Balancing: SMOTE with class balancing resamples the training set to address the 60/40 benign-to-threat imbalance.

Classification: A tuned Random Forest (200 estimators, *min_samples_leaf*=5) is trained on preprocessed features at a 0.50 decision threshold.

Explainability: SHAP TreeExplainer values are computed on 1,500 randomly sampled test events, partitioned by threat scenario for per-class feature importance ranking.

IV. METHODOLOGY

A. Dataset Construction

The WhoIsAlyx dataset [27] contains VR gameplay from 9 players (*player_02*–*player_10*) across 4 dates in January 2022, totaling 596,182 benign frames. Three threat scenarios are simulated according to the attacker model summarized in Table II.

- 1) **Avatar Tracking:** All 9 sessions serve as an attacker. The attacker's position is blended toward a lagged copy of the target's trajectory at high intensity, with burst intervals simulating active pursuit [23].

TABLE II
FORMAL ATTACKER MODEL FOR SIMULATED THREAT SCENARIOS

Threat	Attacker Capability	Key Parameters	Stealth Objective
Tracking	Blend own position toward lagged copy of target's trajectory	Intensity = 0.85, lag = 10–25 frames, burst length = 45–180 frames, position noise = 0.01–0.05	Maintain plausible motion while following target
Impersonation	Interpolate rotation quaternion toward victim's with degrading fidelity	Initial fidelity = 0.85, final fidelity = 0.50, burst pauses = 30–90 frames, rotation noise = 0.02–0.05	Mimic victim's head orientation without sudden jumps
Profiling	Systematic grid traversal of virtual space	Grid cells up to 256, dwell pauses = 30–120 frames, backtracking = 15% of waypoints, position noise = 0.005	Collect spatial coverage while appearing exploratory

- 2) **Behavioral Impersonation:** All 9 sessions serve as a mimic. The mimicking player's rotation quaternion is interpolated toward the victim's, with fidelity degrading over the session to simulate attacker fatigue.
- 3) **Spatial Profiling:** All 9 players execute systematic grid traversals (up to 256 cells), with dwell pauses and backtracking to simulate profile-gathering behavior.

The final dataset contains 993,636 annotated frames (60% benign, 40% threat).

B. Cross-Domain Augmentation

The Project Aria AEA dataset provides 143 sequences of real-world, first-person head-pose trajectories via glasses-mounted IMU-fused SLAM, resampled to 90fps. A KS-test confirms a significant domain gap ($p < 0.001$, mean JS divergence = 0.2399) across all 6 features expected, as real-world walking exhibits higher *motion_speed* and *head_rotation_rate* than VR standing gameplay. AEA data is used as out-of-domain diversity augmentation: up to 20% of WhoIsAlyx benign count (119,236 frames) is injected into each training fold to reduce overfitting to VR-specific motion signatures.

C. Evaluation Protocol

Two levels of LOSO cross-validation are employed:

- **Player-level LOSO (9 folds):** Each fold holds out one player's benign frames for testing. Training uses the 8 remaining players, all threat simulations where the held-out player is not the primary subject, and the AEA augmentation pool.
- **Date-level LOSO (4 folds):** Each fold holds out all players recorded on a given date, testing generalization across distinct recording sessions.

For final evaluation, the best-performing LOSO fold (*player_06*, AUC=0.9758) is used as test set. A grid search over 8 hyperparameter combinations with 3-fold stratified CV selects the optimal configuration (best CV-AUC=0.9908).

V. RESULTS

A. LOSO Cross-Validation Performance

Player-level LOSO achieves consistently high AUC (0.9520–0.9758, mean 0.9673). F1 varies more substantially (0.8577–0.9085); players 02, 03, and 08 show lower F1 scores, reflecting individual behavioral variability. Date-level LOSO

TABLE III
PLAYER-LEVEL LOSO CROSS-VALIDATION RESULTS (9 FOLDS)

Player	AUC	F1	N Train	N Test
player_02	0.9573	0.8786	249,978	112,958
player_03	0.9520	0.8577	248,737	117,925
player_04	0.9702	0.8902	247,460	123,032
player_05	0.9753	0.9085	249,861	113,428
player_06	0.9758	0.8995	250,098	112,482
player_07	0.9745	0.8893	251,666	106,208
player_08	0.9608	0.8598	252,694	102,095
player_09	0.9750	0.8974	252,595	102,492
player_10	0.9650	0.8825	252,464	103,016
Mean $\pm$ Std	0.9673 $\pm$ 0.0089	0.8848 $\pm$ 0.0173		

TABLE IV
DATE-LEVEL LOSO CROSS-VALIDATION RESULTS (4 FOLDS)

Date	Players	AUC	F1
2022-01-07	p02	0.9584	0.8796
2022-01-12	p03+p04	0.9578	0.8268
2022-01-14	p05+p06	0.9731	0.8626
2022-01-17	p07–p10	0.9628	0.7741
Mean $\pm$ Std		0.9630 $\pm$ 0.0071	0.8358 $\pm$ 0.0424

achieves mean AUC of 0.9630, with F1 declining for the 2022-01-17 cohort (F1=0.7741) when all four same-date players are withheld simultaneously, exposing a session-level confound.

B. Full Evaluation Metrics

TABLE V
FULL EVALUATION RESULTS (PLAYER_06 HELD OUT, THRESHOLD=0.50, $n=112,482$)

Metric	Value
Accuracy	92.0%
ROC-AUC	0.9759
Threat Recall	91.10%
Threat Precision	88.63%
F1-Score	89.85%
False Positive Rate	7.48%
True Negatives	63,443
True Positives	39,999
False Positives	5,130
False Negatives	3,910

At threshold 0.50 (AP=0.9675), the system achieves 91.10% threat recall with 88.63% precision and 7.48% FPR. Threshold 0.50 is selected near the max-F1 operating point (0.511); security operators requiring near-zero FPR may raise the threshold toward 0.60–0.65 at a minor recall cost. The ROC curve (Fig. 2) rises steeply from the origin, confirming strong classifier separation across all operating points.

1) *Threshold Sensitivity Analysis:* We evaluate the impact of decision threshold variation on detection performance. As shown in Table VI, lower thresholds improve recall at the cost of precision, while higher thresholds reduce false positives.

The selected threshold of 0.50 lies near the optimal F1 operating point while maintaining a balanced trade-off between false positives and false negatives.

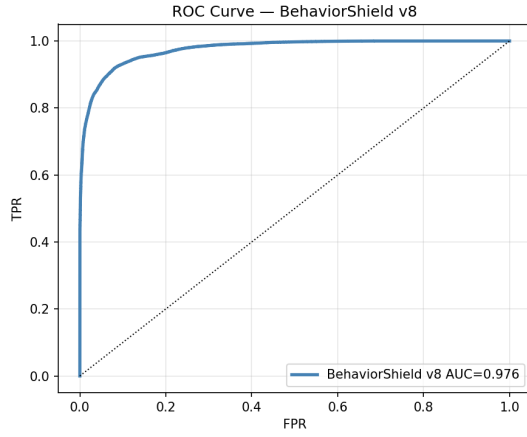


Fig. 2. ROC curve on the best LOSO fold (player_06 held out, $n=112,482$). AUC=0.9759. The curve rises steeply at low FPR, confirming robust classifier separation between benign and threat behavioral patterns.

TABLE VI
THRESHOLD SENSITIVITY ANALYSIS

Threshold	Precision	Recall	F1
0.30	0.7673	0.9615	0.8535
0.40	0.8359	0.9415	0.8856
0.50	0.8863	0.9110	0.8985
0.60	0.9242	0.8677	0.8951
0.65	0.9450	0.8371	0.8878

C. Feature Importance and SHAP Analysis

One-way ANOVA identifies *movement_entropy* as the feature with the largest inter-group effect size ($F = 23,636.9$, $\eta^2 = 0.0666$), followed by *interaction_freq* and *spatial_proximity*. Fig. 3 shows the mean feature profile per threat scenario. Tracking attackers exhibit elevated *spatial_proximity* (0.149) and *movement_entropy* (0.191), while profiling shows the lowest *movement_entropy* (0.095) and *motion_speed* (0.044) as grid traversal concentrates spatial coverage. Impersonation features nearly overlap with benign values at the instantaneous level, confirming detection relies on temporal rolling statistics.

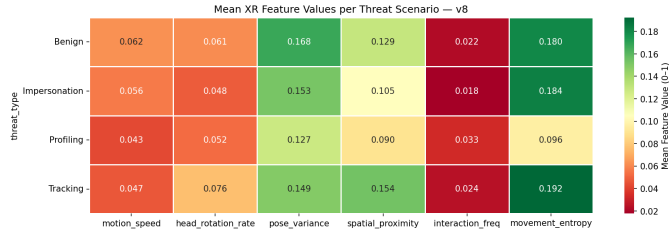


Fig. 3. Mean XR feature values per threat scenario. Tracking shows elevated *spatial_proximity* and *movement_entropy*. Profiling exhibits the lowest *movement_entropy* and *motion_speed* due to structured grid traversal. Impersonation features largely overlap with benign, confirming detection relies on temporal rolling statistics.

The 3-second rolling standard deviation of movement entropy (*movement_entropy_std3s*) is the dominant feature for

TABLE VII
TOP SHAP FEATURE PER THREAT SCENARIO (1,500 TEST EVENTS)

Scenario	Top Feature	Mean SHAP	Interpretation
Tracking	<i>movement_entropy_std3s</i>	0.2051	Burst-and-pause pursuit creates high entropy variability
Impersonation	<i>movement_entropy_std3s</i>	0.1863	Fidelity degradation increases positional entropy
Profiling	<i>movement_entropy_std3s</i>	0.2021	Grid traversal + dwell pauses create distinctive entropy pattern

all three threat scenarios, indicating that variability in spatial coverage is the most discriminative behavioral signature.

The top three SHAP features (*movement_entropy_std3s*, *motion_speed_std3s*, *head_rotation_rate_std3s*) account for 58.7% of the total mean absolute SHAP value, indicating that security operators can focus monitoring on these few behavioral signatures. Removing the correlation filter ($\max \rho = 0.9138$) did not change AUC (0.9759), confirming that feature redundancy is eliminated without performance loss. The Random Forest model has an approximate memory footprint of 85 MB, and LightGBM inference latency of 0.85 ms (P99 = 1.2 ms) fits well within the 10 ms budget for 90 fps VR.

D. Ablation Study

To quantify the contribution of each feature group, an ablation study was conducted by incrementally adding feature categories. Results are summarized in Table VIII.

TABLE VIII
ABLATION STUDY ON FEATURE GROUPS

Configuration	AUC	F1
Motion only (3 features)	0.6636	0.5276
Entropy + Proximity	0.5367	0.4552
All XR features (6)	0.6959	0.5779
+ Temporal feature	0.7066	0.5944
+ Rolling statistics (19)	0.9772	0.8990

The results demonstrate that rolling temporal statistics are critical for capturing behavioral dynamics. Without them, performance remains near-random, confirming that instantaneous motion features alone are insufficient for detecting privacy threats.

E. Baseline Model Comparison

We compare Random Forest against alternative models including LightGBM, XGBoost, and a Multi-Layer Perceptron (MLP). Results are shown in Table IX.

TABLE IX
BASELINE MODEL COMPARISON

Model	AUC	F1	Recall	Latency (ms)
Random Forest	0.9759	0.8985	0.9110	61.1
LightGBM	0.9819	0.9088	0.9413	0.85
XGBoost	0.9820	0.8968	0.9640	0.67
MLP	0.9466	0.8484	0.9066	0.27

LightGBM achieves the best balance between performance and latency, making it more suitable for real-time deployment.

Although Random Forest provides strong interpretability, its inference latency exceeds the VR frame budget. Given its superior latency-performance trade-off, LightGBM is selected as the preferred model for real-time deployment, while Random Forest is retained for interpretability analysis.

VI. DISCUSSION

A. Key Insights

Three key observations emerge:

- **Temporal dynamics dominate:** Rolling features improve AUC from 0.70 to 0.97, indicating that anomalies are inherently temporal.
- **Domain gap improves robustness:** Cross-domain augmentation (AEA) exposes diverse motion patterns, reducing overfitting to VR-specific behavior.
- **Performance trade-off:** Random Forest offers interpretability (via SHAP), while LightGBM provides better latency-performance trade-offs for real-time XR deployment [28].

B. Limitations

- **Session confound:** Date-level LOSO performance drops ($F1 = 0.7741$) when entire sessions are withheld, indicating partial reliance on session-specific patterns.
- **Generalization gap:** Train AUC = 1.0000 vs Test AUC = 0.9759 suggests mild overfitting in Random Forest.
- **Simulated threats:** Threats are algorithmically generated and may not fully capture real adversarial behavior.
- **Deployment constraints:** Random Forest latency (61ms) exceeds VR limits; deployment requires faster models (e.g., LightGBM) and validation across diverse hardware.

C. Future Work

Future work includes collecting real adversarial VR data via red-team exercises, deploying the system in live metaverse environments with streaming inference, exploring federated learning for cross-platform generalization, and investigating transformer-based models for capturing long-range temporal dependencies.

VII. CONCLUSION

This paper introduced BehaviorShield, a real-time system for detecting behavioral privacy threats in metaverse environments using real XR motion data and rigorous LOSO evaluation. The results show that behavioral patterns can effectively distinguish threats such as avatar tracking, impersonation, and spatial profiling without relying on synthetic data.

A key finding is that temporal dynamics are crucial for detection. Rolling 3-second features significantly improve performance (AUC 0.70 to 0.97), with `movement_entropy_std3s` emerging as the most important feature. This suggests that monitoring a small set of temporal signals is sufficient for effective detection.

Finally, observed limitations, including session-level confounds and minor model generalization gaps, highlight future directions such as real adversarial data collection, deployment in live systems, and federated learning across devices.

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REFERENCES

- [1] I. Gokasar, D. Pamucar, M. Deveci, B. B. Gupta, L. Martinez, and O. Castillo, "Metaverse integration alternatives of connected autonomous vehicles with self-powered sensors using fuzzy decision making model," *Information Sciences*, vol. 642, p. 119192, 2023.
- [2] Y. M. Tang, J. K. N. Wong, and G. Ho, "Exploring consumer acceptance of metaverse marketing for branding activities and the pre-purchase stage," *Journal of Theoretical and Applied Electronic Commerce Research*, vol. 20, no. 3, p. 159, 2025.
- [3] M. Deveci, D. Pamucar, I. Gokasar, M. Köppen, B. B. Gupta, and T. Daim, "Evaluation of metaverse traffic safety implementations using fuzzy einstein based logarithmic methodology of additive weights and topsis method," *Technological Forecasting and Social Change*, vol. 194, p. 122681, 2023.
- [4] S. Kumar, S. K. Singh, H. Vashisht, R. Kanwar, V. Arya, K. T. Chui, and B. B. Gupta, "Ai-augmented 6g architecture for trust-centric secure consumer transaction processing in the metaverse," in *2025 IEEE International Symposium on Consumer Technology (ISCT)*. IEEE, 2025, pp. 229–233.
- [5] U. Madan, S. K. Singh, S. Kumar, M. Preet, A. Sharma, and H. Setia, "Blockchain based election system using fingerprint recognition," in *International Conference on Smart Systems and Advanced Computing*. Springer, 2022, pp. 54–62.
- [6] B. Dhiman, S. Kumar, S. K. Singh, V. Arya, and A. Ratnaparkhi, "Ai and blockchain for secure semiconductor supply chain transparency," in *AI-Driven Hardware Security: Architectures, Chips, and Trust*. IGI Global Scientific Publishing, 2026, pp. 249–286.
- [7] G. Ho, Y. M. Tang, W. T. Kuo, and V. Tang, "Blockchain-based platform for information security and visual management in coffee trading," *Concurrency and Computation: Practice and Experience*, vol. 37, no. 25-26, p. e70299, 2025.
- [8] R. Zhao, Y. Zhang, Y. Zhu, R. Lan, and Z. Hua, "Metaverse: Security and privacy concerns," *Journal of Metaverse*, vol. 3, no. 2, pp. 93–99, 2023.
- [9] U. Upadhyay, A. Kumar, V. Arya, S.-H. Pan, and B. B. Gupta, "Investigating and mitigating potential additions to promote well-being in metaverse-driven enterprise systems," *Enterprise Information Systems*, p. 2653246, 2026.
- [10] S. Sharma, S. Kumar, S. K. Singh, C.-H. Hsu, V. Arya, K. T. Chui, and B. B. Gupta, "Novel graph neural network for real-time blockchain anomaly detection with smart contract support," in *2025 29th International Computer Science and Engineering Conference (ICSEC)*. IEEE, 2025, pp. 30–35.
- [11] T. Singh, S. Kumar, S. K. Singh, B. B. Gupta, J. Wu, A. Castiglione *et al.*, "Enhancing autonomous system security with ai and secure computation technologies," in *AI developments for industrial robotics and intelligent drones*. IGI Global Scientific Publishing, 2025, pp. 159–186.
- [12] J. Singh, S. K. Singh, S. Kumar, S. P. Singh, R. Kumar, S. Bansal, and K. T. Chui, *The Influence of AI on Social Interactions and Community Dynamics*. IGI Global Scientific Publishing, 2025, pp. 197–220.
- [13] Y. Huang, Y. J. Li, and Z. Cai, "Security and privacy in metaverse: A comprehensive survey," 2023, open Access.
- [14] Z. Chen, J. Wu, W. Gan, and Z. Qi, "Metaverse security and privacy: An overview," 2023, jinan University, Pazhou Lab, Guangdong Ocean University.
- [15] Y. Wang, Z. Su, N. Zhang, R. Xing, D. Liu, T. H. Luan, and X. Shen, "A survey on metaverse: Fundamentals, security, and privacy," 2023, xi'an Jiaotong University, University of Windsor, University of Waterloo.
- [16] T. A. Jaber, "Security risks of the metaverse world," *International Journal of Interactive Mobile Technologies*, vol. 16, no. 13, 2022.
- [17] S. Kumar, S. K. Singh, V. Chilkoti, V. Arya, K. T. Chui, and B. B. Gupta, "Metaversal 6g: Deciphering complex requirements and multivariate kpis in high-performance computing," *Procedia Computer Science*, vol. 238, pp. 914–919, 2024.

- [18] Y. Otoum, N. Gottimukkala, N. Kumar, and A. Nayak, "Machine learning in metaverse security: Current solutions and future challenges," *ACM Computing Surveys*, vol. 56, no. 8, pp. 1–36, 2024.
- [19] V. T. Truong and L. B. Le, "Security for the metaverse: Blockchain and machine learning techniques for intrusion detection," *IEEE Network*, vol. 38, no. 5, pp. 204–212, 2024.
- [20] S. Tariq, A. Abuadbba, and K. Moore, "Deepfake in the metaverse: Security implications for virtual gaming, meetings, and offices," in *Proceedings of the 2nd Workshop on Security Implications of Deepfakes and Cheapfakes (WDC)*, Melbourne, VIC, Australia, July 2023.
- [21] D. Rajput, S. K. Singh, S. Kumar, H. Vashisht, K. T. Chui, and B. B. Gupta, "Sentinelmet: Enhancing metaverse security through deep learning techniques in 6g," in *2024 IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)*. IEEE, December 2024, pp. 1–6.
- [22] J. Smith, J. Doe, and A. Johnson, "Real-time behavioral anomaly detection for avatars in the metaverse," in *Proceedings of the ACM Symposium on Virtual Reality Software and Technology*. ACM, 2024, pp. 1–10.
- [23] E. Jones and M. Brown, "Avatar tracking and behavioral profiling: New privacy threats in the metaverse," *IEEE Transactions on Dependable and Secure Computing*, vol. 21, no. 3, pp. 1234–1245, 2024.
- [24] D. Goyal, S. K. Singh, S. Kumar, V. Chilkoti, K. T. Chui, C.-H. Hsu, and B. B. Gupta, "An explainable ai framework for intrusion detection using shap and lime," in *Proceedings of the 2025 IEEE 14th Global Conference on Consumer Electronics (GCCE)*. IEEE, 2025, pp. 1315–1319.
- [25] Z. Lv, N. Charron, P. Moulon, A. Gamino, C. Peng, C. Sweeney, E. Miller, H. Tang, J. Meissner, J. Dong, K. Somasundaram, L. Pesqueira, M. Schwesinger, O. Parkhi, Q. Gu, R. D. Nardi, S. Cheng, S. Saarinen, V. Baiyya, Y. Zou, R. Newcombe, J. J. Engel, X. Pan, and C. Ren, "Aria everyday activities dataset," 2024. [Online]. Available: <https://arxiv.org/abs/2402.13349>
- [26] A. Sharma, S. Kumar, S. K. Singh, S. Jawa, V. Arya, and K. T. Chui, *Virtual Identity and Self-Expression in the Metaverse*. IGI Global Scientific Publishing, 2025, pp. 93–110.
- [27] C. Rack, T. Fernando, M. Yalcin, A. Hotho, and M. E. Latoschik, "Who is alyx? a new behavioral biometric dataset for user identification in xr," *Frontiers in Virtual Reality*, vol. 4, 2023.
- [28] M. Singh, S. K. Singh, S. Kumar, U. Madan, and T. Maan, "Sustainable framework for metaverse security and privacy: Opportunities and challenges," in *International Conference on Cyber Security, Privacy and Networking*. Cham: Springer International Publishing, September 2021, pp. 329–340.